



Agentic AI Community of Practice

DAAI CoP

Journey — 7th June 2026



02

Udemy Based Learning

Focused Agentic AI training through structured Udemy courses



DAAI Agentic AI CoP Overview

ENTRY LEVEL

DAAI Agentic AI Foundation

- ChatGPT Python Programming: AI Guided Code Mastery
- Data Engineering 101: The Beginner's Guide
- Intro to AI: A Beginner's Guide to Artificial Intelligence
- Generative AI for Beginners
- Intro to AI Agents and Agentic AI
- Agentic AI for Beginner: Build AI Agents From Scratch

INTERMEDIATE

DAAI Agentic AI Advance

- ChatGPT Python Programming: AI Guided Code Mastery
- AI Engineer Core Track: LLM Engineering, RAG, QLoRA, Agents
- The Agentic AI Engineering Masterclass 2026

ADVANCED

DAAI Agentic AI Expert

- ChatGPT Python Programming: AI Guided Code Mastery
- AI Engineer Core Track: LLM Engineering, RAG, QLoRA, Agents
- AI Engineer Agentic Track: The Complete Agent & MCP Course

CoP Evaluation Framework

01

Udemy Training

- ▶ Focused Agentic AI training courses on Udemy
- ▶ Tracking with L&D team support

02

Capstone Project

- ▶ Tailored Assignments – Capstone projects across domains for different roles
- ▶ Levels: Agentic AI Foundation, Agentic AI Advance & Agentic AI Expert

03

Assessment

- ▶ Assessment via PeopleRise – Learning & Development Suite
- ▶ Evaluation based on tailored criteria & metrics for all roles
- ▶ Learning recommendation post assessment

Udemy Enrolment

- 1 Login to Udemy
<https://apexon.udemy.com/>
- 2 Select My Learning → Learning Path
- 3 Enrol as per your assigned CoP (see right)
- 4 Complete all courses at the earliest
- 5 If more than one CoP assigned in learning path, report back

CoP Learning Paths

DAAI Agentic AI Expert

Agentic AI Engineering Expert: Python, LLMs & AI Agents

DAAI Agentic AI Advance

Agentic AI Advance: Engineering & LLM Development

DAAI Agentic AI Foundation

Agentic AI Foundation: AI, Data, and Autonomous Systems

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Capstone Project

Real-world agentic AI projects across domains — build, deploy, and demonstrate your expertise.



Capstone Projects — Agentic AI Advance CoP

Agentic AI Advance Requirements

CP01

Enterprise Knowledge Assistant

Multi-agent RAG system answering employee queries with citations. | Tools: LangChain/LangGraph, Vector DB, LLM

CP03

Autonomous Meeting Action Agent

Extracts decisions and action items from meeting transcripts. | Tools: Speech-to-Text, LLM, NER, Calendar API

CP05

Certification Evaluation Agent

Evaluates assignments using structured rubrics and confidence scoring. | Tools: LLM, Structured Output Parser

CP07

Automated Loan Approval Agent

Evaluates loan applications with risk scoring and explainability. | Tools: ML Models, Policy Engine, SHAP, LLM

CP09

Predictive Maintenance Agent

Forecasts equipment failure using sensor data analysis. | Tools: Time-Series Model, LLM, Dashboard

CP02

Agentic Email Triage System

Classifies emails, drafts responses, and escalates based on rules. | Tools: Email API, LLM, LangGraph

CP04

Personalized Learning Agent

Adapts educational content dynamically based on learner progress. | Tools: Vector DB, LLM, User State Store

CP06

IT / Ticket Auto-Resolution Agent

Analyzes IT tickets and recommends or executes resolution steps. | Tools: ServiceNow API, LLM, RAG

CP08

Claims Processing Optimization Agent

Flags anomalies in insurance claims and recommends settlements. | Tools: Anomaly Detection, Graph Analytics, LLM

CP10

Dynamic Pricing Agent

Recommends optimal pricing using demand and constraint modeling. | Tools: XGBoost, Bandits, Optimization Engine

Capstone Projects — Agentic AI Expert CoP

CP11

Supply Chain Disruption Management Agent

Detects supply disruptions and simulates mitigation strategies.

CP12

Autonomous Product Recommendation Agent

Balances personalization with inventory and margin constraints.

CP13

Energy Consumption Optimization Agent

Analyzes energy usage and recommends cost-saving optimizations.

CP14

Business Process Optimization Agent

Discovers bottlenecks from process logs and suggests improvements.

CP15

Clinical Decision Support Agent

Provides evidence-based recommendations with safety checks.

CP16

Regulatory Compliance Monitoring Agent

Continuously monitors policy and operational data for compliance gaps.

CP17

Contract Review & Negotiation Agent

Analyzes legal contracts and suggests risk-aware negotiation changes.

CP18

Autonomous Incident Response Agent

Detects system anomalies and proposes safe remediation steps.

CP19

Prompt Injection & AI Security Agent

Red-teams AI systems to detect prompt injection and vulnerabilities.

CP20

Self-Healing AI System

Monitors AI performance and triggers safe remediation for failures.

CP21

Executive Insight / Strategy Agent

Synthesizes multi-source business data into executive insights.

CP22

Smart City Operations Agent

Optimizes traffic, utilities, and public systems using real-time data.

Agentic AI Expert Requirements

Capstone Projects — Agentic AI Foundation CoP

CP23

Intelligent Personal Finance Advisor Agent

Fully functional Agentic AI system that acts as an Intelligent Personal Finance Advisor. The agent autonomously reads financial data, reasons over it, takes multi-step actions, and produces actionable personal finance recommendations.

Tools Required: Python, LangChain, Pandas, LLM API

Task lists

- ▶ Complete all assigned Foundation courses
- ▶ Build CP23 as your capstone
- ▶ Use open-source LLM API provided
- ▶ Generate synthetic financial data
- ▶ Present working demo as PPT

Agentic AI Foundation Requirements

Tools Details

Code	Capstone Project Name	CoP	Tools Required
CP01	Enterprise Knowledge Assistant	Agentic AI Advance	LangChain/LangGraph, Vector DB, LLM, FastAPI/Streamlit
CP02	Agentic Email Triage System	Agentic AI Advance	Email API, LLM, LangGraph, Rule Engine
CP03	Autonomous Meeting Action Agent	Agentic AI Advance	Speech-to-Text, LLM, NER, Calendar API
CP04	Personalized Learning Agent	Agentic AI Advance	Vector DB, LLM, User State Store
CP05	Certification Evaluation Agent	Agentic AI Advance	LLM, Structured Output Parser
CP06	IT / Ticket Auto-Resolution Agent	Agentic AI Advance	ServiceNow API, LLM, RAG, Logging System
CP07	Automated Loan Approval Agent	Agentic AI Advance	ML Models, Policy Engine, SHAP, LLM
CP08	Claims Processing Optimization Agent	Agentic AI Advance	Anomaly Detection, Graph Analytics, LLM
CP09	Predictive Maintenance Agent	Agentic AI Advance	Time-Series Model, LLM, Dashboard
CP10	Dynamic Pricing Agent	Agentic AI Advance	XGBoost, Bandits, Optimization Engine
CP11	Supply Chain Disruption Management Agent	Agentic AI Expert	NetworkX, Simulation Engine, LLM
CP12	Autonomous Product Recommendation Agent	Agentic AI Expert	Collaborative Filtering, Bandits, LLM
CP13	Energy Consumption Optimization Agent	Agentic AI Expert	Time-Series Models, Optimization Solver
CP14	Business Process Optimization Agent	Agentic AI Expert	Process Mining (PM4Py), LLM
CP15	Clinical Decision Support Agent	Agentic AI Expert	Medical RAG, LLM, Guardrails
CP16	Regulatory Compliance Monitoring Agent	Agentic AI Expert	Policy Engine, Log Ingestion, Risk Scoring Model
CP17	Contract Review & Negotiation Agent	Agentic AI Expert	Legal RAG, Clause Classifier, LLM
CP18	Autonomous Incident Response Agent	Agentic AI Expert	Observability Tools, LLM, Safe Execution Sandbox
CP19	Prompt Injection & AI Security Agent	Agentic AI Expert	Adversarial Prompt Set, LLM Evaluator, Risk Classifier
CP20	Self-Healing AI System	Agentic AI Expert	Drift Detection, Eval Framework, Agent Orchestrator
CP21	Executive Insight / Strategy Agent	Agentic AI Expert	Data Pipelines, KPI Engine, LLM
CP22	Smart City Operations Agent	Agentic AI Expert	IoT Ingestion, Optimization Engine, Multi-Agent Planner
CP23	Intelligent Personal Finance Advisor Agent	Agentic AI Foundation	Python, Langchain, Pandas, LLM API

LLM API Details

Below is the sample Python code for querying the LLM using a local LLaMA training model.

```
# streaming response
import requests
import json

url = "http://3.94.255.185:11434/api/generate"
payload = {
    "model": "llama3.2",
    "prompt": "Explain me Schrodinger's cat paradox in simple terms.",
    "stream": True
}

with requests.post(url, json=payload, stream=True) as response:
    response.raise_for_status()
    for line in response.iter_lines():
        if line:
            data = json.loads(line.decode("utf-8"))
            if "response" in data:
                print(data["response"], end="", flush=True)
            if data.get("done"):
                break
```

Sample Single Agent Code

Single-Agent script using LangChain + Ollama | Save as First_Agent.py

Part 1 of 2 ▶

```
from typing import Any
from langchain_core.tools import tool
from langchain_ollama import ChatOllama
from langchain.agents import create_agent
from ddgs import DDGS

# 1. Initialize the local model using the remote server address
llm = ChatOllama(base_url="http://3.94.255.185:11434", model="llama3.2", temperature=0)

# 2. Define the web search tool using ddgs directly
@tool
def duckduckgo_search(query: Any) -> str:
    """Search the web using DuckDuckGo. Useful for current information."""
    if isinstance(query, dict):
        query_str = None
        for key in ["value", "query", "q", "input", "text"]:
            if key in query:
                query_str = str(query[key])
                break
    if query_str is None:
        str_vals = [str(v) for v in query.values() if isinstance(v, (str, int, float))]
        query_str = str_vals[0] if str_vals else str(query)
```

Sample Single Agent Code - Contd..

Continued from previous slide | First_Agent.py

◀ Part 2 of 2

```
else:
    query_str = str(query)

print(f"\n[Tool Executing] DuckDuckGo search for: '{query_str}'")
with DDGS() as ddgs:
    results = list(ddgs.text(query_str, max_results=3))
    output = "\n".join(r["body"] for r in results) if results else "No results found."
    print(f"[Tool Result] Found {len(results)} results.")
    return output

tools = [duckduckgo_search]

# 3. Create the agent (LangChain 1.3.4 API)
agent = create_agent(
    model=llm,
    tools=tools,
    system_prompt="You are a software developer. Use duckduckgo_search to find Agentic AI certifications."
)

# 4. Run the agent
response = agent.invoke({"messages": [{"role": "user", "content": "Share details about AWS Certifications"}]})
print("\n--- FINAL ANSWER ---")
print(response["messages"][-1].content)
```

High Level Activities

Project Work

- ▶ Select Capstone project from the available project list
- ▶ Use requirement document as guideline; apply your domain knowledge to improve
- ▶ Use Apexon-provided laptop for training and Capstone project
- ▶ Use Open Source LLM API key provided — Free & Unlimited Tokens
- ▶ For any licensed tool, find an equivalent open-source alternative
- ▶ Use CoPilot or any suitable AI tool to generate required code
- ▶ Prepare PPT for final demo

Data & Resources

Synthetic Data

Generate Synthetic data for input files

Kaggle Datasets

<https://www.kaggle.com/datasets>

UC Irvine ML

<https://archive.ics.uci.edu/datasets>

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Assessment

Via PeopleRise — Learning & Development Suite



Assessment — Step by Step

01

Login to PeopleRise

<https://peoplerise.apexon.ai/login>

Use your Apexon credentials

02

L&D Suite App

Navigate to Learning & Development Suite App

Launch Agent

03

Take the Assessment

Expert CoP → select AI Architect role

Advance & Foundation CoP → select AI Engineer role

Select suitable role

04

Get Certified

The agent will provide your score along with personalised UdeMy course recommendations to support your learning in the field of AI. You may enrol in any suggested course to enhance your knowledge.

Certification is awarded on successful completion of assessment



Thank You

DAAI Agentic AI Community of Practice

Let's build the future of Agentic AI — together.

